WORKED SOLUTION

STEP III 2019, Question 9

Mechanics • Particle on Moving Hemisphere

STEP III 2019, Question 9

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

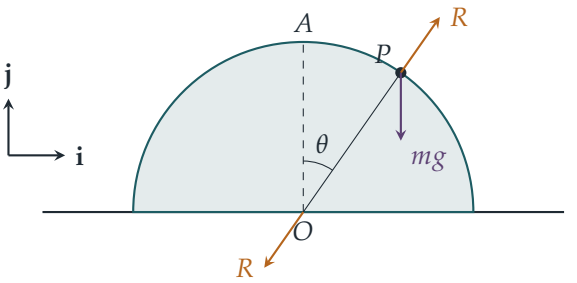
www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Both surfaces are smooth, so no external force on the system (particle plus hemisphere) has a horizontal component: horizontal momentum is conserved, and that ties the hemisphere's speed to the particle's angular speed. Since nothing does work except gravity, energy is conserved too, and those two laws give part (ii). For leaving the surface, the clean test is that the normal reaction becomes zero; the suggested vector is chosen so that the unknown $\ddot{\theta}$ disappears when we take a component.

Solution

Model: P is a particle, the hemisphere is rigid, and all contacts are smooth. While they touch, the only force between P and the hemisphere is a normal reaction R along OP, and the table pushes the hemisphere vertically.



The reaction on the hemisphere (drawn at O for clarity) is equal and opposite to the one on P.

Part (i)

At time t the centre O has moved to $-s\,\mathbf{i}$, and P is at distance a from it in direction $\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}$. So

$$\mathbf{r} = (a\sin\theta - s)\,\mathbf{i} + a\cos\theta\,\mathbf{j},$$

and differentiating with respect to t (chain rule on $\sin\theta$ and $\cos\theta$) gives

$$\dot{\mathbf{r}} = (a\dot\theta\cos\theta - \dot s)\,\mathbf{i} - a\dot\theta\sin\theta\,\mathbf{j},$$

as required.

The external forces on the system are the two weights and the table's reaction, all vertical, so the total horizontal momentum stays at its initial value, which is zero (we take the small initial displacement to give negligible speed). The hemisphere's velocity is $-\dot s\,\mathbf{i}$, so

$$m(a\dot\theta\cos\theta - \dot s) - M\dot s = 0 \quad\Longrightarrow\quad \dot s = \frac{m}{m+M}\,a\dot\theta\cos\theta = (1-k)\,a\dot\theta\cos\theta,$$

because $1 - k = \dfrac{m}{m+M}$. Substituting into the $\mathbf{i}$-component of $\dot{\mathbf{r}}$,

$$a\dot\theta\cos\theta - (1-k)a\dot\theta\cos\theta = k\,a\dot\theta\cos\theta, \qquad \text{so} \qquad \dot{\mathbf{r}} = a\dot\theta\,(k\cos\theta\,\mathbf{i} - \sin\theta\,\mathbf{j}).$$

Part (ii)

The reactions do no net work: the table's reaction is perpendicular to the hemisphere's motion, and the pair of forces $\pm R$ between P and the hemisphere do no total work because P stays on the surface (their relative velocity is along the surface, perpendicular to R). So kinetic plus potential energy is conserved. P has dropped by $a(1-\cos\theta)$, and initially everything is (essentially) at rest, so

$$\tfrac12 ma^2\dot\theta^2(k^2\cos^2\theta + \sin^2\theta) + \tfrac12 M\dot s^2 = mga(1-\cos\theta).$$

Using part (i), $M\dot s^2 = M(1-k)^2a^2\dot\theta^2\cos^2\theta$. The total coefficient of $a^2\dot\theta^2\cos^2\theta$ is then

$$mk^2 + M(1-k)^2 = \frac{mM^2 + Mm^2}{(m+M)^2} = \frac{mM}{m+M} = mk.$$

Hence $\tfrac12 ma^2\dot\theta^2(k\cos^2\theta + \sin^2\theta) = mga(1-\cos\theta)$, and dividing by $\tfrac12 ma$,

$$a\dot\theta^2(k\cos^2\theta + \sin^2\theta) = 2g(1-\cos\theta).$$

Part (iii)

Write $\mathbf{u} = \sin\theta\,\mathbf{i} + k\cos\theta\,\mathbf{j}$. Differentiating the last result of part (i),

$$\ddot{\mathbf{r}} = a\ddot\theta\,(k\cos\theta\,\mathbf{i} - \sin\theta\,\mathbf{j}) - a\dot\theta^2\,(k\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}).$$

The first bracket is perpendicular to $\mathbf{u}$, since $k\cos\theta\sin\theta - \sin\theta\,k\cos\theta = 0$. So

$$\ddot{\mathbf{r}}\cdot\mathbf{u} = -a\dot\theta^2(k\sin^2\theta + k\cos^2\theta) = -k\,a\dot\theta^2.$$

Ask yourself Why is this particular vector useful?

It kills the $\ddot{\theta}$ term, which we know nothing about, leaving only $\dot{\theta}^2$, which part (ii) gives us. Now compare with Newton's second law for P:

$$m\ddot{\mathbf{r}} = -mg\,\mathbf{j} + R(\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}).$$

Taking the component along $\mathbf{u}$ and equating with the result above,

$$-mk\,a\dot{\theta}^2 = -mgk\cos\theta + R(\sin^2\theta + k\cos^2\theta),$$

so

$$R(\sin^2\theta + k\cos^2\theta) = mk(g\cos\theta - a\dot{\theta}^2).$$

The particle leaves the hemisphere when R falls to zero. Since $k > 0$, at $\theta = \alpha$ this gives

$$a\dot{\theta}^2 = g\cos\alpha.$$

The cubic. Put $c = \cos\alpha$ and use part (ii) at $\theta = \alpha$, with $\sin^2\alpha = 1 - c^2$:

$$gc(kc^2 + 1 - c^2) = 2g(1 - c) \quad\Longrightarrow\quad (1-k)\,c^3 - 3c + 2 = 0.$$

Let $f(x) = (1-k)x^3 - 3x + 2$, where $0 < 1 - k < 1$ because m and M are both positive.

- $f'(x) = 3\big((1-k)x^2 - 1\big) < 0$ for $-1 \le x \le 1$, so f is strictly decreasing there and has at most one root in $[-1, 1]$.
- $f\left(\frac{2}{3}\right) = \frac{8}{27}(1-k) - 2 + 2 = \frac{8}{27}(1-k) > 0$ and $f(1) = -k < 0$.

So the only root in $[-1, 1]$ lies strictly between $\frac{2}{3}$ and 1. As $\cos\alpha$ must lie in $[-1, 1]$, it is this root, and

$$\cos\alpha > \tfrac{2}{3}.$$

(This is consistent with the formula for R: substituting part (ii) shows $g\cos\theta - a\dot{\theta}^2$ has the sign of $-f(\cos\theta)$, which is positive at $\theta = 0$ and first vanishes at this root, so P really does stay in contact until then.)

Checks

- If $M \to \infty$ then $k \to 1$, the hemisphere stays put, and the cubic becomes $-3c + 2 = 0$: the familiar $\cos\alpha = \frac{2}{3}$ for a particle sliding off a fixed sphere. The moving hemisphere makes the particle leave earlier, at a larger $\cos\alpha$.
- For equal masses, $1 - k = \frac{1}{2}$ and the cubic $c^3 - 6c + 4 = 0$ factorises as $(c-2)(c^2 + 2c - 2) = 0$, giving $\cos\alpha = \sqrt{3} - 1 \approx 0.732$, which exceeds $\frac{2}{3}$.
- A step-by-step numerical simulation of the full motion by computer, for several mass ratios, confirmed the relations in parts (i) and (ii) along the way and found the reaction vanishing at $\cos\theta \approx 0.732$ when $m = M$, matching the cubic.

Common slips

- Using the fixed-sphere condition $a\dot{\theta}^2 = g\cos\theta$ without justification: here the hemisphere accelerates, so the particle's acceleration is not just circular motion about a fixed point. It happens to hold at the instant of leaving only because $R = 0$ means the hemisphere has no horizontal acceleration at that moment.
- Forgetting the hemisphere's kinetic energy $\frac{1}{2}M\dot{s}^2$ in the energy equation.
- Showing only that a root lies between $\frac{2}{3}$ and 1, without ruling out another root in $[-1, 1]$.

Beyond STEP

This is a two-degree-of-freedom system, and Lagrangian mechanics handles it systematically: write the kinetic and potential energies in terms of θ and s, and the conservation of horizontal momentum appears automatically because s does not appear in the Lagrangian (an "ignorable" coordinate, linked to a symmetry by Noether's theorem). Constraint forces such as R are found afterwards, just as we did here. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to build the Newtonian intuition first.

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